



Mid-Holocene soil water and vegetation in the Xi'an area of the southern Chinese Loess Plateau

Da-Peng Yue^a, Jing-Bo Zhao^{b,c,*}, Yan-Dong Ma^{d,*}, Jun-Ji Cao^c, Yan-Hua Li^a, Ai-Hua Ma^a, Xiu-Ming Liu^e

^a School of Geography and Tourism, Shaanxi Normal University, Xi'an 710119, Shaanxi, China

^b State Key Laboratory of Loess and Quaternary Geology, Institute of Earth Environment, Chinese Academy of Sciences, Xi'an 710061, Shaanxi, China

^c Key Laboratory of Aerosol Chemistry and Physics, Institute of Earth Environment, Chinese Academy of Sciences, Xi'an 710061, Shaanxi, China

^d Key Laboratory of Subsurface Hydrology and Ecological Effects in Arid Region, Ministry of Education, Chang'an University, Xi'an 710054, Shaanxi, China

^e School of Geography, Fujian Normal University, Fuzhou 350000, Fujian, China

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ABSTRACT

This paper investigates mid-Holocene palaeo-vegetation and soil water in the Xi'an area of the southern Chinese Loess Plateau (CLP). A well-dated (8.5–6.0 ka BP and 5.0–3.1 ka BP) loess-palaeosol (unit S₀) was logged, sampled and analyzed at four sites in the Xi'an area of the southern CLP, from which we report soil micro-morphological (type and occurrence of root pores), particle size and geochemical data (CaCO₃ and Sr concentration) to: (i) infer changes in soil water content, (ii) infer the limits of gravity water and the types of hydrologic cycle, and (iii) reconstruct the palaeovegetation. Numerous woody plant root holes were identified (2–4 mm diameter) within the illuvial horizon underlying palaeosol S₀, the morphology of which is consistent with those produced by woody plants. The clay-rich argillic horizon of palaeosol S₀ was found to contain high concentrations of kaolinite, which in concert with the geochemical indices, indicates that soil gravity water attained a depth of 2.2–2.8 m, whilst soil water content was ~20–25%. This implies that pedogenic processes at this site were more intense during the mid- compared to late Holocene loessic soils elsewhere on the CLP, and ultimately attests to relatively wet climatic conditions (e.g. up to 800 mm yr⁻¹ rainfall), the hydrologic cycle type of precipitation-soil water-groundwater, and the vegetation of deciduous broad-leaved forest. We propose that combined analyses of root-channel morphology and mobile element geochemistry can be used to infer changes in palaeo-hydrology and ultimately, the strength of the Asian Summer Monsoon.

1. Introduction

Palaeoenvironmental changes during the Holocene are a major focus of research because of their impacts on human population and well-being (Bond et al., 1997; Matsumoto, 2005; Wang et al., 2010). Notably, the Holocene is characterised by a number of abrupt climate changes, which are manifested in various palaeoenvironmental archives (Denton and Karlen, 1973; Chen et al., 2016; Hong et al., 2005). Pollen records from lake sediments reveal five major phases of climate change during the Holocene in the monsoon region of east China (Chen et al., 1978), which is consistent with pollen records in Europe (Wendland and Bryson, 1974). Lake sediment records in both northwest and southwest

China show that there were around 10 abrupt climate events (Chen et al., 2001; Zhang et al., 2008; Wang and Feng, 2013; Leipe et al., 2014), which correlate with the Greenland ice core record (O'Brien et al., 1995). Stalagmite $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ values also provide evidence for abrupt Holocene climate changes in the monsoon region of China; but with the caveat that the $\delta^{18}\text{O}$ record is sensitive to precipitation not temperature and, as such, indicate only 3–4, rather than 10 abrupt climatic shifts (Wang et al., 2005, 2019a; Hang et al., 2017). Such abrupt climatic changes are also recorded within the loess-palaeosol sequences of the Chinese Loess Plateau (CLP) (Sun and Zhao, 1991; Shi et al., 1992). In particular, an extensive mid-Holocene palaeosol (S₀) dated to 8500–3100 a BP (An et al., 1985; Sun and Zhao, 1991; Guo and Liu,

* Corresponding authors at: State Key Laboratory of Loess and Quaternary Geology, Institute of Earth Environment, Chinese Academy of Sciences, Xi'an 710061, Shaanxi, China (Jing-Bo Zhao); Key Laboratory of Subsurface Hydrology and Ecological Effects in Arid Region, Ministry of Education, Chang'an University, Xi'an 710054, Shaanxi, China (Yan-Dong Ma).

E-mail addresses: zhaojb@snnu.edu.cn (J.-B. Zhao), mayandongemail@163.com (Y.-D. Ma).

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1996; Huang et al., 2002, 2006; Wang et al., 2014; Mao et al., 2017; Pang et al., 2017), has been interpreted as evidence for moister climate conditions relative to the early- and late Holocene across the CLP. Deposition of a 15 cm thick loess layer within this palaeosol between 6000 and 5000 a BP evidences an abrupt shift to cold conditions during which the rate of aeolian deposition overtook pedogenesis, but is only preserved at a few sites in China (Huang et al., 2002, 2006; Lei et al., 2019). However, evidence for a similar cold period has been discovered in the North Caucasus Mountains (Alexandrovskiy, 2000; Munoz et al., 2017). In addition, tree rings provide the highest resolution records of climate change down to annual and even seasonal scales (Yang et al., 2000; Liu et al., 2005; Wang et al., 2019b), but a disadvantage of these records is that they only provide relatively short term records (i.e. up to 3 ka).

Previous studies across the CLP have shown that favorable climate conditions during the mid-Holocene were instrumental in the development of agricultural production and agricultural technology (Huang et al., 2002; Pang et al., 2017; Cui et al., 2018), key to which was the soil water status. However, lack of quantitative data pertaining to soil water and its relations with palaeoprecipitation during the mid-Holocene, impedes our understanding of the operational limits of the prevailing ecosystems and the implications for the sustainability of agricultural activities.

In this paper, several Holocene loess-palaeosol profiles were investigated with a view to reconstructing the soil water and vegetation type which prevailed during the formation of palaeosol S_0 . Recognition of such changes is of great importance if we are to further understand the impacts of Holocene climate change on human populations and refine predictive models of the future impacts of climate change on food production and water resource availability across the CLP.

2. Study area, materials and methods

The Xi'an area is currently subjected to a temperate monsoon climate with an mean annual temperature of 13.2 °C and a mean annual precipitation of 600–700 mm (Song, 1987). Fully developed, continuous loess strata which record palaeo-environmental changes from the Early Pleistocene to the Holocene, persist in this area and attain thicknesses of up to 120 m. The Holocene loess strata can be divided into 2 layers: (i) late Holocene loess (L_0) up to 1.0 m thick; (ii) the mid-Holocene

palaeosol (S_0) up to 0.8 m thick. This mid-Holocene palaeosol (S_0) is underlain by the 6 m thick Malan loess (L_1) unit. The boundary age between units S_0 and L_1 is 8500 years; whilst the boundary age between units S_0 and L_0 is 3100 years (Huang et al., 2006, 2012; Pang et al., 2017).

Based on field investigations, four Holocene loess profiles in the Xi'an area (Wujiafen, Ancun, Xiwangcun and Sifucun), were selected in this study (Fig. 1). The Wujiafen profile is located on the second terrace of the south bank of the Wei River (400 masl), 4 km south of Xi'an city center. The Sifucun profile at Chang'an is located in the middle part of the Shaoling tableland (600 masl) about 12 km south of Xi'an city center. The Ancun profile is located at the eastern end of the Bailu tableland, (700 masl) 6 km southwest of Lantian County. The Xiwangcun profile is located in the southeast of the Bailu tableland (700 masl) about 14 km southwest of Lantian town. At all four profiles, a weathered, leached loess layer was developed below unit S_0 whilst the $CaCO_3$ illuvial horizon within unit S_0 was also distributed into the upper part of unit L_1 . Therefore, units L_0 , S_0 , and the upper part of L_1 were sampled in this study.

Each profile was sampled at 5 cm intervals to determine $CaCO_3$ and Sr content as well as undertake particle size analyses. A total of 460 samples were collected from the 4 profiles. The gas volume method was applied to analyze $CaCO_3$ content. Soil mineral composition was determined by the X-ray diffraction (XRD) method. The $CaCO_3$ migration depth was determined using measured $CaCO_3$ concentrations. The soil microstructure was identified using an optical microscope. The Sr content was determined using an X-ray elemental analyzer. Particle size analysis was undertaken using laser diffraction methods. The distribution of plant root holes and root diameter were then measured. All measurements were completed at Shaanxi Normal University.

3. Results

3.1. Profile stratifications within the mid-Holocene palaeosol

The mid-Holocene palaeosol S_0 in Xi'an is well-developed and can be classified as a dark loessic soil. The Holocene and late Pleistocene loess strata exposed at the four sites can be divided into five horizons (Figs. 2 and 3). The thicknesses and pedological characteristics of each unit are summarised in Table 1 and outlined in the following text: (i) Unit L_1 is 6



Fig. 1. The location of Xi'an and studied sections.

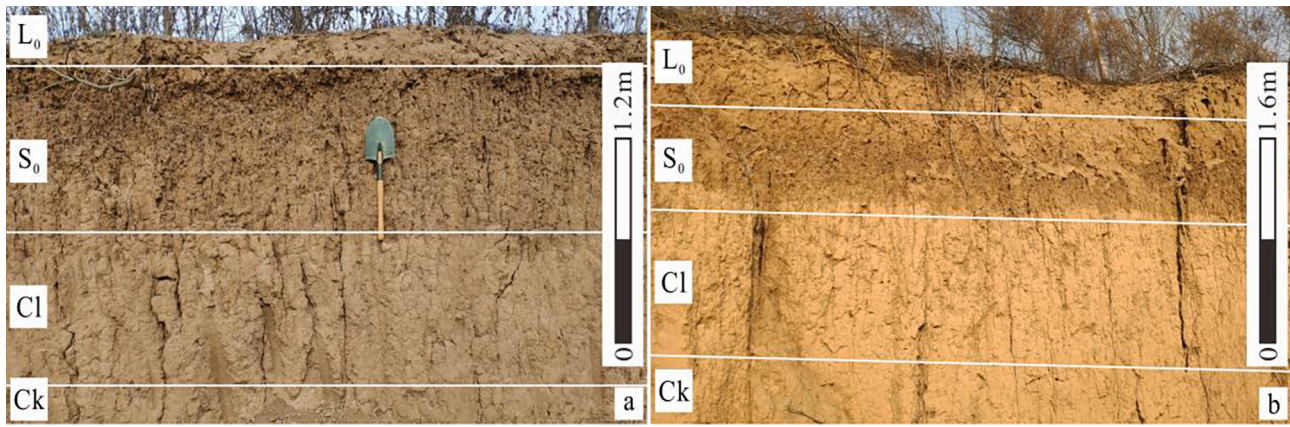


Fig. 2. Holocene loess profile in Xiwangcun (a) and Ancun (b) of the Xi'an area. L₀, Holocene loess; S₀, Mid-Holocene palaeosol; Cl, The weathered and leached loess layer; Ck, the CaCO₃ illuvial horizon of the S₀.

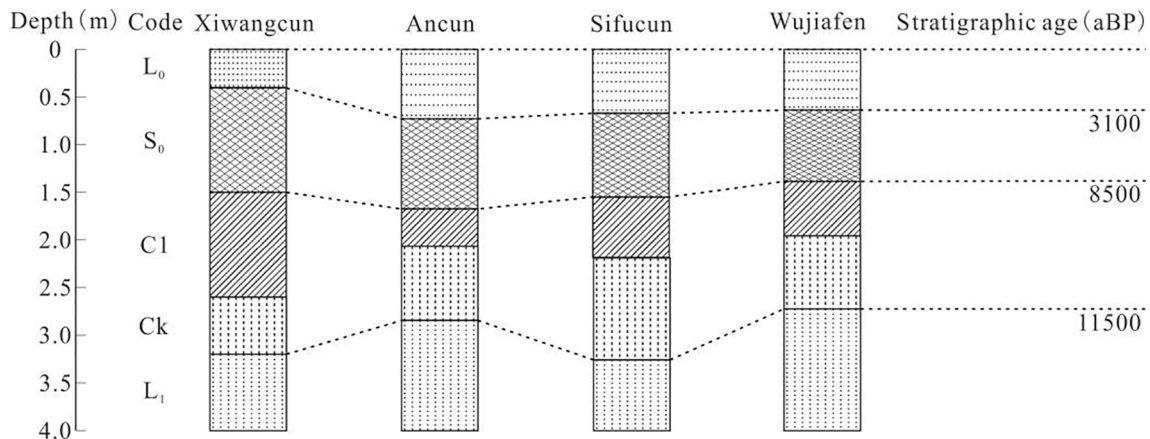


Fig. 3. Lithostratigraphy of loess sequences investigated in this study, dating of boundary layers after Huang et al. (2002, 2006) and Pang et al. (2017). L₀, Holocene loess; S₀, Mid-Holocene palaeosol; Cl, The weathered and leached loess layer; Ck, the CaCO₃ illuvial horizon of the S₀; L₁, first loess or Malan loess.

Table 1

Thickness of every horizon in Holocene palaeosol (m).

Horizon	Bt	Cl	Ck	Horizon	Bt	Cl	Ck
Ancun	0.92	0.40	0.80	Sifucun	0.90	0.64	1.20
Xiwangcun	1.10	1.10	0.60	Wujiafen	0.80	0.60	1.10

m thick and is friable with a granular structure. Colour is grayish yellow; (ii) Unit Ck is a 0.6–1.2 m thick CaCO₃ illuvial horizon of palaeosol S₀, which consists of CaCO₃ concretions and white films that are distributed in the lower part of the early Holocene loess; (iii) Unit Cl consists of accretions of weathering products derived from heavy leaching of palaeosol S₀, and also includes the leached early Holocene loess (0.6–1.1 m thick), which is grayish yellow, and exhibits a granular and agglomerate structure; (iv) the fourth unit (palaeosol S₀) is an argillic horizon (0.8–1.1 m thick) which is dark grayish brown and exhibits a coarse prismatic structure; (v) whilst unit L₀ (0.6–1.1 m thick), is grayish yellow and exhibits a friable, granular structure. The upper part of this unit has been disturbed by human activities. Summarily, the Holocene loess strata in these profiles can be classified as Bt(S₀)-Cl-Ck-C(L₁) soil horizons, respectively. In particular, CaCO₃ migration was related to soil palaeo-water content. Thus, the evidence for intensive leaching of the loess (below unit S₀) reflected in the CaCO₃ illuvial horizon may be important with respect to linking soil palaeo-water content with climate.

At all four sites, palaeosol S₀ is punctuated by a thin loess layer which was deposited at ~6000–5000 a BP and corresponds with the timing of

loess deposition more widely across the CLP (Huang et al., 2002, 2006; Lei et al., 2019). Since this loess unit has been pedogenically modified, it is generally considered to be part of palaeosol S₀ (An et al., 1985; Sun and Zhao, 1991; Guo and Liu, 1996; Wang et al., 2014). Previous studies have demonstrated that loess accretion in this period occurred in response to cold, dry climatic conditions (Huang et al., 2002, 2006; Lei et al., 2019).

3.2. Root holes' of woody plants in the mid-Holocene palaeosol and late-Holocene loess

Root holes associated with woody plants are distributed from the lower boundary of palaeosol S₀ to a depth of 2 m within unit L₁. Notably, the distribution density of root holes exhibits a gradual decline from top (5–10 cm intervals) to bottom (20–50 cm intervals). The cross-sectional diameter of the root holes is typically 1–3 cm, but several range from 3 to 5 cm. The morphology of the root holes is typically circular, the longitudinal orientation is duct-like, relatively flat and exhibits a vertical distribution (Fig. 4a-c). Dark gray to dark brown illuviated clays are present in most of the root 'holes'. The colour and particle size of these illuviated clay fillings are the same as those contained in the S₀ argillic horizon.

3.3. CaCO₃ content and migration depth in the mid-Holocene palaeosol

CaCO₃ content in both the S₀ argillic horizon and the leached loess unit is low (average less than 0.61%, Fig. 5). CaCO₃ content is

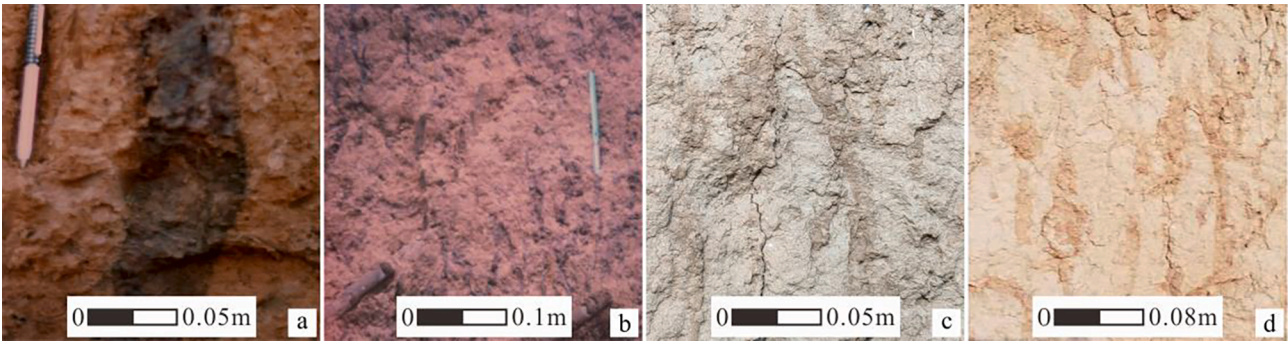


Fig. 4. Root holes of trees or brushes with fillings in loess profiles from the Xi'an study site: a. Root holes of trees or brushes with fillings in loess profiles from the Xi'an study site: a. The root holes filled with S_0 palaeosol material in the L_1 loess and CI beneath the S_0 palaeosol Wujiafen; b. Dense root holes filled with groundmass from the S_0 palaeosol in the L_1 loess and CI beneath the S_0 palaeosol at Wujiafen; c. Dense root holes filled with groundmass from the S_0 palaeosol material in the L_1 loess and CI beneath the S_0 palaeosol at Ancun; d. Dense root holes filled with groundmass from the S_0 palaeosol material in the L_1 loess and CI beneath the S_0 palaeosol at Xiwangcun.

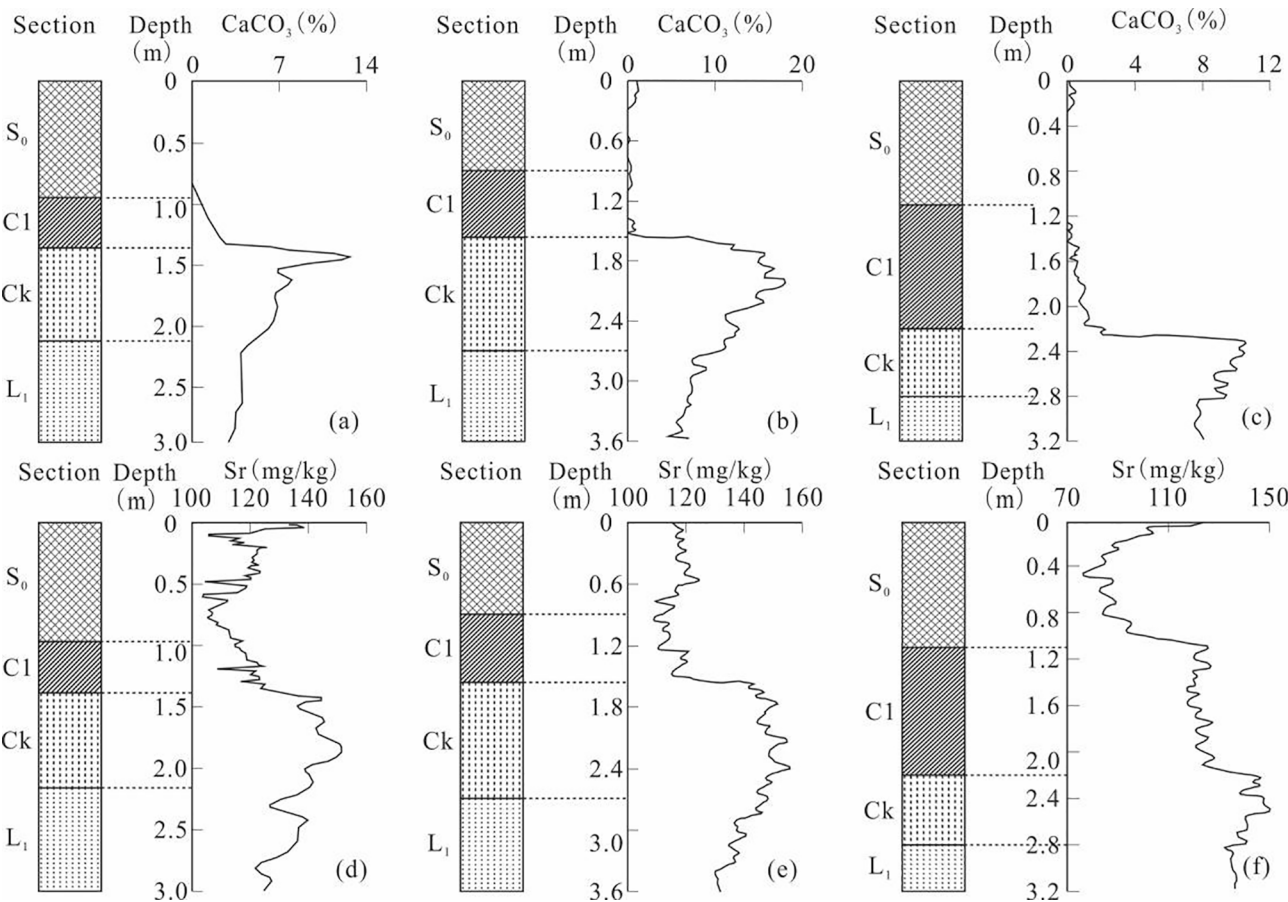


Fig. 5. $CaCO_3$ and Sr concentration profiles of loess from the Xi'an region. $CaCO_3$ and Sr concentration profiles of loess from the Xi'an region. $CaCO_3$ concentrations calculated on the basis of carbonate analyses. (a) Ancun, (b) Sifucun, (c) Xiwangcun.

concentrated in nodules and is high in the illuvial horizon (average 7.10%) (Fig. 5). The average $CaCO_3$ content of the parent material (L_1) beneath the $CaCO_3$ illuvial horizon is 3.57% (Fig. 5). Notably, the $CaCO_3$ illuvial horizon within this unit is distributed from 1.4 to 2.2 m depth (average 1.8 m) similar to the Wujiafen profile (average 1.9 m). The $CaCO_3$ content within the S_0 unit exposed at both the Sifucun and Xiwangcun profiles is similar to that within the Ancun profile (Fig. 5, Table 2). However, the $CaCO_3$ cements have accumulated at greater depths at these profiles (2.2 and 2.5 m depth respectively) compared to the Ancun and Wujiafen profiles.

Table 2				
CaCO ₃ content and migration depth in Wujiafen profile.				
Horizon	Bt horizon	CI horizon	Ck horizon	L ₁ horizon
CaCO ₃ content range (%)	0.00–2.43	0.00–3.17	8.71–16.34	5.39–9.87
Mean CaCO ₃ content (%)	0.52	1.33	12.51	7.32
Depth in each horizon (m)	0–0.80	0.80–1.40	1.4–2.5	2.5–3.3

3.4. Sr content and mobility within the mid-Holocene palaeosol

Sr content is very low within palaeosol S_0 and the leached loess unit (average 115.3 and 120.8 mg/kg respectively) (Fig. 5). In contrast the illuvial horizon enriched in CaCO_3 nodules contains comparatively high Sr content (average 142.8 mg/kg) (Fig. 5). Furthermore, Sr concentration is higher in the upper part of unit L_1 below the CaCO_3 illuvial horizon, but lower than the CaCO_3 illuvial horizon (average 131.1 mg/kg) (Fig. 5). The average depth of Sr accumulation within unit S_0 is 1.8 m (range 1.4–2.2 m). The Sr content of each layer within palaeosol S_0 is similar in all profiles except Xiwangcun. Given that Xiwangcun is proximal to the Qinling Mountains, the most likely explanation for this difference is that higher rainfall resulted in the loss of Sr from this profile leading to its depletion relative to the other sites.

3.5. Microstructure of the mid-Holocene palaeosol

Twenty samples from the S_0 argillic horizon and 7 root casts were identified and analyzed under a microscope (Fig. 6). Results reveal that silt is the dominant grain size in the S_0 argillic horizon with subordinate clay content and a poorly developed pore structure. Quartz dominates the soil mineralogy. Dark heavy minerals are also present but highly weathered. In particular, the microstructure is characterized by a significant number of neo-argillans, which occur in the form of a flow gel and block (Fig. 6a–e). Argillans can be divided into two types: (i) residual argillans which are distributed at the edge of the soil aggregate (Fig. 6b and c), the clay particles which lack any distinct orientation; and (ii) illuviated argillans formed by the migration and accumulation of clay, which has produced clearly orientated clay particles (Fig. 6a, d and e). The microstructure of the root hole fillings is characterized by a uniform structure and the argillans are sometimes observed along the edges of root cast fillings. No debris structure is observed.

3.6. Mineralogical composition of the mid-Holocene palaeosol

XRD analyses were conducted on 10 samples from the S_0 argillic horizon, the results of which are summarized in Table 3. The XRD

spectra are dominated by quartz (50%) followed by feldspar (15%), illite-montmorillonite (15%), chlorite (6%), kaolinite (4%) and calcite (3%). In the Malan loess unit (L_1) beneath palaeosol S_0 , calcite and feldspar concentrations are relatively high compared with the argillic horizon of palaeosol S_0 , but the concentrations of quartz, illite-montmorillonite and kaolinite are low.

3.7. Particle size distribution of the mid-Holocene palaeosol

Clay content is indicative of the degree of pedogenesis within loess-palaeosol sequences. The results of particle size analyses from each of the different sites is outlined in turn.

Within the Ancun profile, the clay content of the argillic horizon within palaeosol S_0 is relatively high, ranging from 20.3% to 30.7%, (average 23.9%) (Fig. 7). The content of clay in the weathered and leached loess layer is similar to that of the argillic horizon ranging from 20.4 to 24.2% (average 22.5%) (Fig. 7). The clay content within the illuvial horizon enriched in CaCO_3 nodules is relatively low, ranging from 14 to 24% (average 18.1%) (Fig. 7). In the upper part of unit L_1 below the CaCO_3 illuvial horizon, clay content is very similar to the CaCO_3 illuvial horizon, ranging from 15.3 to 21% (average 18.6%) (Fig. 7). Both palaeosol S_0 and the weathered and leached loess unit have undergone significant clayification caused by pedogenesis. This is reflected in the fact that coarse silt and clay content are inversely correlated (Fig. 7a). Were the clay to have been allochthonous, both silt and clay fractions could be expected to correlate, which is not the case.

Within the Xiwangcun profile, the clay content in palaeosol S_0 ranges from 18.8 to 29.2%, which is higher compared to the Ancun profile (average 24.8%) (Fig. 7). Conversely, the clay content within the weathered and leached loess unit is low relative to the Ancun profile, ranging from 15.4 to 20.9% (average 17.7%) (Fig. 7). Notably the upper part of the weathered and leached loess layer contains higher clay content than the lower part (Fig. 7). Clay concentration within the CaCO_3 nodule illuvial horizon is also low, ranging from 16.6 to 19.4% (average 17.6%) (Fig. 7). In the upper part of unit L_1 below the CaCO_3 illuvial horizon, the clay content is very low, ranging from 15.8 to 17.3% (average 16.8%) (Fig. 7). Similar to the Ancun profile, the degree of clay

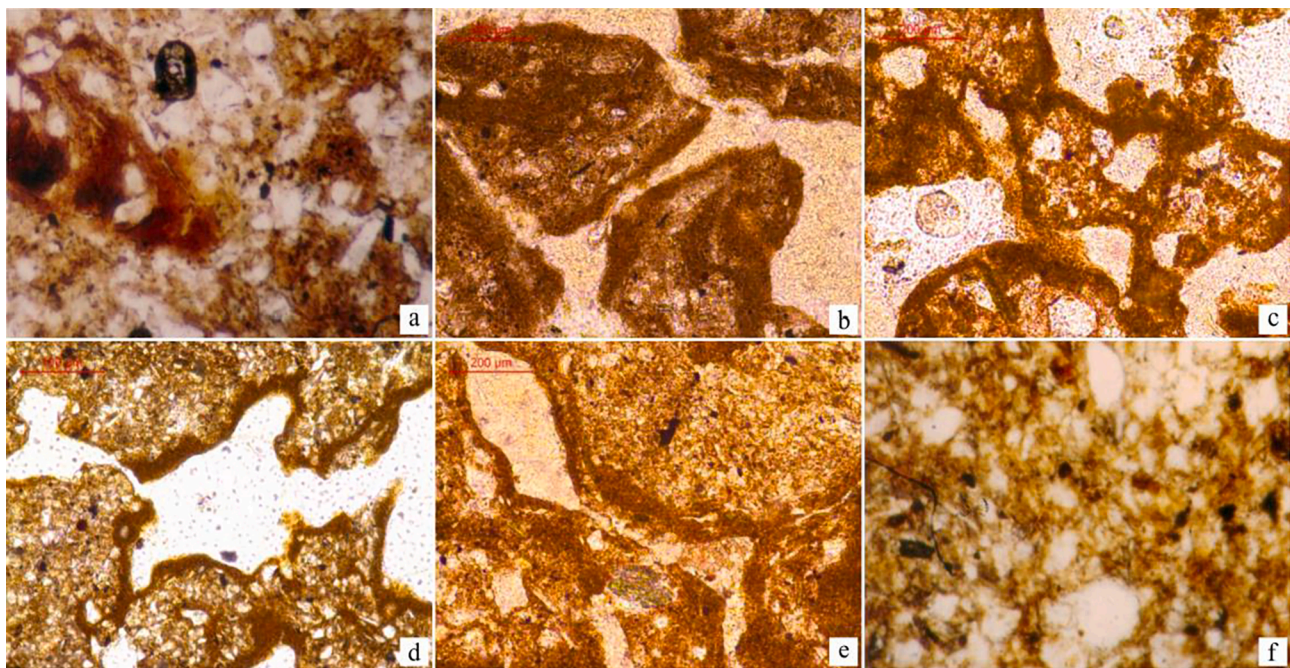


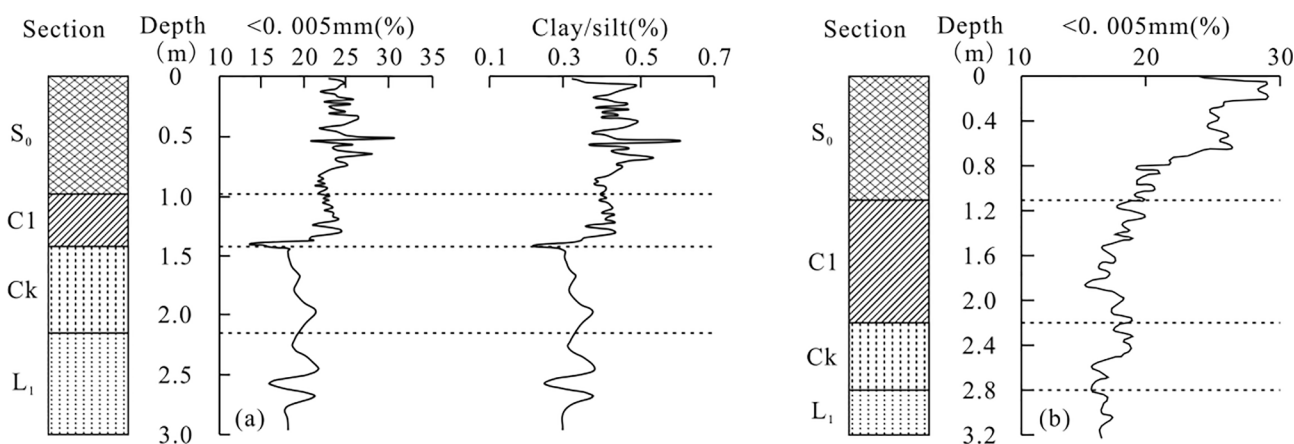
Fig. 6. Microtexture of mid-Holocene palaeosol. a, b. the argillans at the edge of the agglomerate within palaeosol S_0 at Xi'an Wujiafen; c, d. the argillans at the edge of the mass within palaeosol S_0 at Lantian Ancun; e. the argillans at the edge of the fracture zone within palaeosol S_0 at Chang'an and Sifucun; f. the microstructure of clay cementation in palaeosol S_0 at Chang'an and Sifucun.

Table 3

Mineral composition (%) in mid-Holocene palaeosol and the earth filled in root holes in Wujiafen, Xi'an.

Layer	Depth(m)	Quartz	Feldspar	IMON mixed layer	Kaolinite	Chlorite	Calcite	Tremolit	Other minerals
Palaeosol	0.2	54	12	15	4	6	3	–	4
Palaeosol	0.4	55	15	14	3	7	3	–	3
Palaeosol	0.6	54	13	16	4	6	–	3	4
Filling earth	1.2	52	14	15	3	6	5	2	3
Filling earth	1.5	52	16	16	4	4	2	–	6
Filling earth	1.8	50	15	15	5	5	4	3	3
L ₁ loess	2.6	48	20	13	–	8	8	–	3
L ₁ loess	3.3	46	20	12	–	9	9	2	2

Note: IMON represents illite-montmorillonite; the depth is measured from the top of the palaeosol.

**Fig. 7.** Grain-size distribution of the mid-Holocene palaeosol at Ancun (a) and Xiwangcun (b) loess profiles from the Xi'an region.

formation is related to pedogenesis, rather than allochthonous inputs. Thus, the highest clay concentration within the upper part of the weathered and leached loess unit attests to intense pedogenesis.

4. Discussion

4.1. The influence of monsoon activity on mid-Holocene palaeosol development

The CaCO_3 content of the mid-Holocene soils at our study sites is less than 1%, whilst the intermediate depth of CaCO_3 and Sr accumulation is typically 1.8–2.5 m indicative of leaching (Xiong and Li, 1987; ISSS Working Group Reference Base, 1998; Zhu and He, 2001). The occurrence of well-developed argillans in concert with newly formed clay minerals (up to 6%) including kaolinite relative to the weathered and leached parent materials beneath palaeosol S_0 , indicates that palaeosol S_0 is a heavily leached soil equivalent to the 'luvisol' soil type (ISSS Working Group Reference Base, 1998). Notably, pedogenic intensity within this mid-Holocene palaeosol S_0 is shown to be markedly greater than that recorded in the modern drab soils in the Xi'an area (Xiong and Li, 1987; Zhu and He, 2001).

The mid-Holocene (excluding the dry-cool period of 6000–5000a BP) was a period of climatic warming (Huang et al., 2002; Sycheva et al., 2003) during which the temperature was $\sim 2^\circ\text{C}$ higher than present (Sun and Zhao, 1991; Shi et al., 1992). However, significant spatial differences in precipitation influenced the development of palaeosol S_0 across northern China. An et al. (1985) for instance proposed that the average annual precipitation was between 300 and 500 mm at Lantian. Based on the greater depth at which CaCO_3 and Sr are concentrated, in concert with the occurrence of kaolinite clay compared to the modern soils across the Xi'an area (Table 4, Fig. 5, Fig. 8), we suggest that the average annual precipitation during the mid-Holocene, must have been ~ 800 mm (Velde, 1992; Robert and Kennet, 1997).

Precipitation in the Xi'an area is supplied primarily by water vapor

Table 4Migration depth of CaCO_3 and thickness of leaching parent matter layer under S_0 palaeosol (m).

Soils	Sa	Sb	S ₀ a	S ₀ b	S ₀ c	S ₀ d	Sc
Maximum depth of CaCO_3 migration (m)	1.5	1.4	2.2	2.8	2.7	2.5	3.7
Average depth of CaCO_3 migration (m)	1.2	1.1	1.8	2.5	2.1	2.0	3.5
Thickness of leaching parent matter (m)	0.1	0.1	0.5	1.1	0.9	0.6	2.4
Mean annual precipitation (mm)	600	600	750	850	800	780	1050

Note: Sa is Changan modern drab soil; Sb is Zhouzhi modern drab soil; S₀a is Ancun S_0 palaeosol; S₀b is Xiwangcun S_0 palaeosol; S₀c is Sifucun S_0 palaeosol; S₀d is Wujiafen S_0 palaeosol; Sc is the modern subtropical yellow-brown soil of Liuhe, Jiangsu.

from the Asian summer monsoon, but in the present day is blocked by the Qinling Mountains. This implies that the monsoon was strong enough during the mid-Holocene to extend northwards of the Qinling Mountains, and drive soil formation in the Xi'an area.

4.2. Soil water during the formation of palaeosol S_0 in the Xi'an area

Previous studies have suggested that adsorbed water would lead to the dissolution of soluble salts, but their solubility was shown to be weak (Huang et al., 1998). Our observations show that adsorbed water easily dissolved soluble salts, but had no effect on the insoluble carbonates; nor did it regulate the precipitation of CaCO_3 (Zhao, 2004; Zhao et al., 2015). CaCO_3 is a poorly soluble carbonate that requires gravity water for its dissolution (Zhao, 2004; Zhao et al., 2015). For example, after palaeosol layers S_1 , S_4 and S_5 were formed on the Loess Plateau, they were subjected to the effects of water adsorption over timescales of 100,000–500,000 years; yet the CaCO_3 content was still maintained at

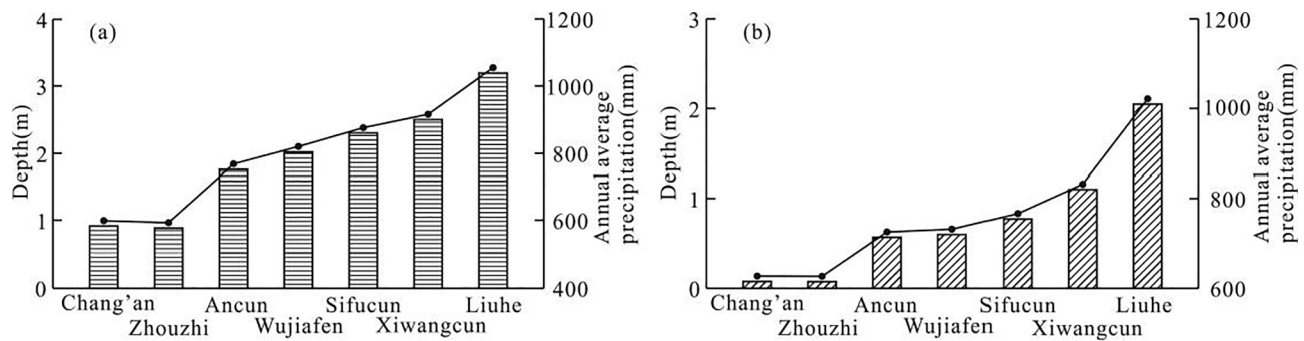


Fig. 8. Estimates of average annual palaeo-precipitation during the mid-Holocene (ca. 8500–3100 a BP). (a) Estimates based on the average depth of migration of CaCO₃ and Sr; (b) estimates based on the depth of the loess sections containing weathered and leached parent material.

or close to 0 (Guo and Liu, 1996; Guo et al., 1998; Zhao, 2004; Zhao et al., 2015) and the CaCO₃ enriched illuvial horizons within the palaeosol were well-preserved (Guo and Liu, 1996; Guo et al., 1998; Zhao, 2004; Zhao et al., 2015). In the modern soils of the Loess Plateau, adsorbed water can reach groundwater level at a depth of ~50 m after many years of migration, but CaCO₃ sedimentation occurs at a depth of less than 2 m within the gravity water belt (Xiong and Li, 1987; Zhang et al., 2017). There is a functional relationship between the migration depth of soil CaCO₃ and annual precipitation from which (palaeo) precipitation can be inferred (Zhao, 2004; Retallack, 2005). The soil water concentration of CO₂, pH, Ca, bicarbonate and the infiltration rate control the solubility of CaCO₃ in seasonally formed gravity water. The soil water budget decreases without continued precipitation and leaves adsorbed water behind. Due to evaporative enrichment and the long interaction time with coexisting soil particles, CaCO₃ becomes oversaturated with CaCO₃ in the pore water fluids. Thus, we interpret the CaCO₃ profile as broadly indicative of the maximum migration depth of gravity water. During the mid-Holocene, gravity water depth was thus about ~2.0 m near Xi'an, but attained a depth of 2.8 m at the Xiwangcun profile near the Qinling Mountains.

The formation of the argillic horizon during the mid-Holocene (Fig. 6), comprised of about 3% kaolinite clay (Table 3) would have required that the groundmass be nearly saturated by soil water (Guo and Liu, 1996; Zhao et al., 2015). The porosity of the late Pleistocene loess (L₁) and the Holocene loess (L₀) is about 50% (Ma et al., 2018), which would in turn imply a field capacity of around 20% (Yang and Shao, 2000). The argillans within palaeosol S₀ are not well-developed, which implies that the soil water content during the formation of the argillans should be slightly higher than the field capacity, yet lower than the saturated water content. When the soil water content is higher than the field capacity, part of the water in the soil exists in the form of gravity water. Therefore, it can be inferred that the soil water content during the development of the mid-Holocene soil would have exceeded 20%, but during the rainy season, soil water content was probably at least 25–30%.

The degree of pedogenic clay mineral formation within palaeosol S₀ is not limited to the argillic horizon itself, as part of the parent material (L₁) below is also affected by clayification (Fig. 7). The depth of clay formation at the Ancun and Xiwangcun profiles attained 1.4 m (Fig. 7a) and 1.5 m respectively (Fig. 7b). Clayification requires high gravity water content to facilitate CaCO₃ dissolution; thus clay formation indicates that the weathered and leached parent material beneath palaeosol S₀ possessed high water content in the rainy season (~20–25%).

Soil material which filled in the root holes of woody plants during the development of palaeosol S₀ also indicates that the soil gravity water during the mid-Holocene attained depths of 2–3 m. The root hole fillings are inherited from the overlying grey brown S₀ argillic horizon (Fig. 4), the filling structure is uniform and dense and does not contain soil debris particles >0.5 mm. This evidences slow percolation typical of gravity water, rather than high infiltration rates. The characteristics and

distribution depth of these dense root hole fillings indicate that the soil above 2–3 m depth was inundated frequently by gravity water during the rainy seasons resulting in soil water content > 20%.

The proposed annual precipitation of 800 mm during the mid-Holocene would have offset soil water loss to evaporation during the dry season. For example, in 2003 after a year of drought, the annual precipitation in Xi'an attained >800 mm. Even after a hot summer the soil water content of the broad-leaf forest in the autumn and winter (at 2 m depth) was still 18–20%; but attained 20–25% at 2–4 m depth (Chen et al., 2006; Du and Zhao, 2006). Therefore, assuming that a drought year has not occurred before a wet year, the soil water content of the broad-leaf forest in the autumn and winter should be about 25%, and even larger during the rainy season and summer. This in concert with the fact that the particle size of palaeosol S₀ is less permeable than that of the modern soil, reinforces our interpretation that a soil water content of 20–25% persisted during the mid-Holocene.

4.3. Hydrologic cycle during the development of the mid-Holocene palaeosol

In general, the types of hydrologic cycle within thick soils are recognised, namely precipitation-soil water and precipitation-soil water-groundwater (Yang and Shao, 2000; Chen et al., 2006; Zhao et al., 2007, 2011). Elemental leaching in the upper soil horizons represents unequivocal evidence for the frequent occurrence of soil gravity water infiltration in these loess sections (Fig. 9). This suggests that the migration depth of CaCO₃ and Sr in these sections with the soil gravity water was about 2.2–2.8 m (Fig. 9). In these loess sections, the root holes of woody vegetation were mainly distributed within a depth of about 2 m (Fig. 4). Although the root system of woody vegetation can reach a depth of greater than 2 m (Li et al., 1993), there are few root systems of woody vegetation below a depth of 2 m. The root system of woody vegetation mainly absorbs and uses soil water within 2 m. The depth at which these leached elements have accumulated in the loess profiles was higher than the distribution depth of the main roots of woody vegetation, which implies that during the formation of palaeosol S₀, soil gravity water below 2 m depth would not have been influenced by evapotranspiration. In the modern forests of the Loess Plateau, a negative soil water flux has been recorded in years when annual precipitation is less than 600 mm, and vice versa (Yang and Shao, 2000). Since the annual precipitation was ~800 mm during palaeosol S₀ formation, which is approximately 200 mm higher than that of modern precipitation, this implies that the hydrologic cycle type in this area was precipitation-soil water-groundwater. The resultant water surplus would have led to excess discharge as groundwater.

Although the amount of gravity water within the loess soils has been shown to >20%, an equally important condition for the maintenance of a positive soil water flux is the infiltration depth. A number of studies have shown that an infiltration depth of 1.5 m is needed to ensure that a positive soil water flux is maintained (Xiong and Li, 1987; Zhu and He,

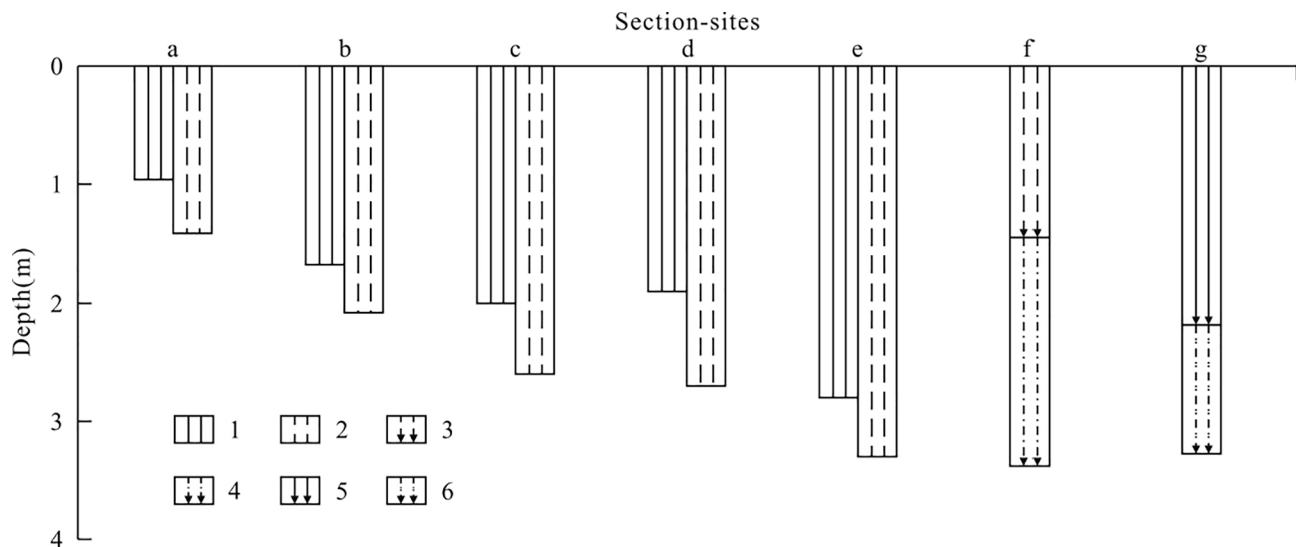


Fig. 9. The depth of gravity water distribution during the development of palaeosol S_0 indicated by the intermediate depth of CaCO_3 and Sr. a. Xi'an modern soil; b. Ancun profile; c. Wujiafen profile; d. Sifucun profile; e. Xiwangcun profile; f. distribution of modern soils in the Xi'an area; g. Xi'an area S_0 water belt in development 1. The intermediate depth of migration of CaCO_3 and Sr; 2. The depth of gravity water distribution; 3. The distribution of gravity water in the modern soils across the Xi'an region; 4. The distribution of water in Xi'an modern soil; 5. The distribution of gravity water during palaeosol S_0 formation; 6. Adsorption water distribution during palaeosol S_0 formation.

2001; Yang and Shao, 2000). For example, the modern annual precipitation of 600 mm in Xi'an has resulted in a gravity water depth of 1.4 m as reflected in the CaCO_3 illuvial horizon (Figs. 8 and 9). Thus at 1.4 m, the soil water would have been in a balanced state. Since the depth of gravity water during the development of palaeosol S_0 was about 2.0 m in the Xi'an area, it follows that soil water flux must have been positive during the mid-Holocene. In addition, there is almost no weathered parent material in the modern soil of this area (Fig. 9), whereas the weathered parent material of palaeosol S_0 reached 30 cm thickness. The weathered parent material of palaeosol S_0 largely can also indicate a positive soil water flux.

4.4. Vegetation during the development of mid-Holocene palaeosol

Previous research has demonstrated that root holes with a diameter of >2 cm at a depth of about 2 m are formed by trees (Li et al., 1993), and are present between 2 and 3 m within palaeosol S_0 at our study sites (Fig. 4). A soil water content $\geq 12\%$ in silt dominated loess coupled with annual precipitation of ~ 800 mm is sufficient to sustain temperate deciduous broad-leaved forests in the modern temperate and subtropical plain areas of China (Xiong and Li, 1987; Yang and Shao, 2000; Zhu and He, 2001; Yang et al., 2000; Zhao et al., 2007; Liu et al., 2010). We therefore propose that deciduous forests were extensive during the mid-Holocene across the CLP.

5. Conclusion

Four loess-palaeosol profiles in the Xi'an area of the southern Loess Plateau were investigated in order to reconstruct palaeo-vegetation and soil water content during the development of mid-Holocene palaeosol S_0 . This was achieved via soil microstructural analyses of plant root pores, supported by measurements of particle size and geochemistry (CaCO_3 , Sr concentrations). This palaeosol is comprised of two layers across this area, which formed around 8500–6000 a BP and 5000–3100 a BP, respectively. The palaeosol is comprised of four soil horizons: argillic, leaching, illuvial and parent material. The results demonstrate that: 1) numerous plant root holes (diameter: 2–4 mm) indicate the development of woody plants during the development of palaeosol S_0 ; 2) CaCO_3 and Sr concentration in the illuvial horizon (e.g. 2.2–2.8 m depth) was higher than for the argillic horizon; 3) new argillans and

kaolinite were formed during palaeosol development; and 4) the clay content in the argillic horizon is significantly higher than the other three layers. Compared with soil water and precipitation during the late Holocene in the Xi'an area, the strong pedogenic overprinting of palaeosol S_0 indicates that: 1) moister conditions (up to 800 mm yr^{-1} rainfall) prevailed, which in turn suggests that the summer monsoon was strong enough to have penetrated the Qinling Mountains and deliver precipitation to the southern CLP; 2) soil gravity water was present around 2.2–2.8 m and soil water content in the rainy season was 20%–25%, indicating that the hydrologic cycle type was 'precipitation-soil water-groundwater' and favored extensive woody plant growth in the study area, possibly broad-leaved deciduous forest. This study demonstrates that CaCO_3 and Sr concentrations within palaeosol S_0 in concert with the thickness of the weathered parent material layer, constitute useful proxies of Holocene soil water balance and hydrologic cycle, which ultimately reveal changes in the strength of the Asian Summer Monsoon.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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